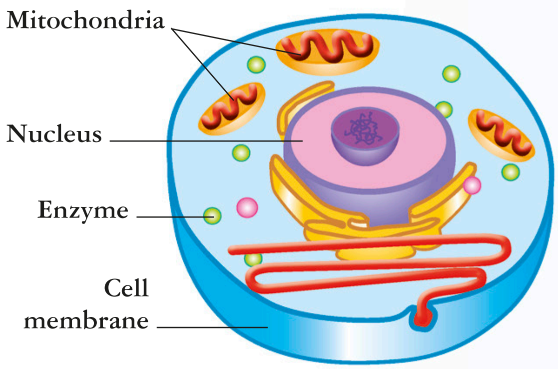
The factory INSIDE a CELL

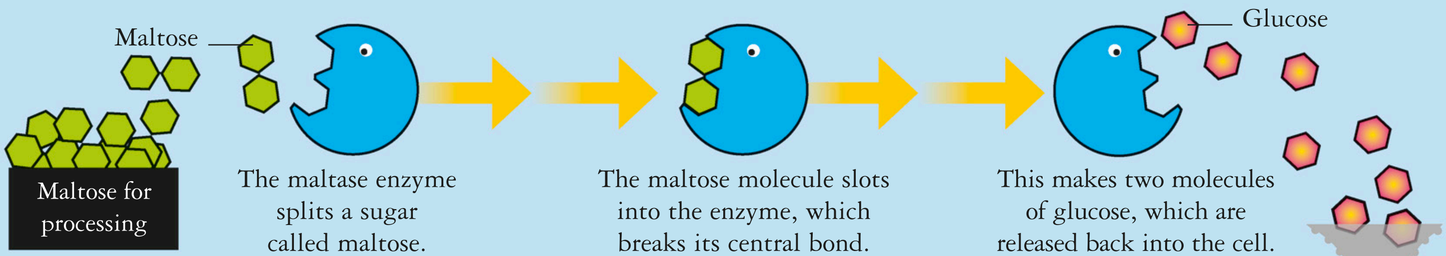


Like tiny factories, every cell in your body is a hive of activity. *Thousands of chemical reactions* are taking place there every second, providing you with energy, building your body, and allowing you to breathe, move, and think.

CELL MEMBRANE

FOOD-PROCESSING DEPOT

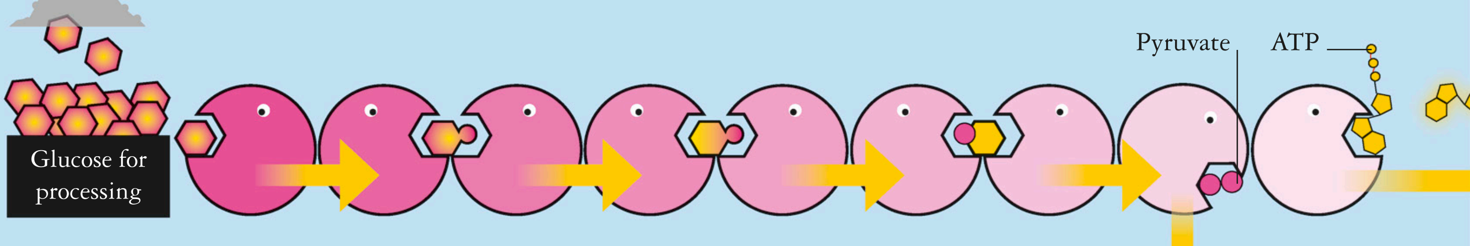
ENZYMES are the workhorses of a cell. Even a bacterium has around 1,000 different enzymes floating inside it, busily carrying out chemical reactions that split or join molecules together. Enzymes are proteins and each folds up into a unique shape. That shape allows it to carry out one specific reaction and do it quickly and efficiently. Enzymes are named after the chemicals they process. This one is called maltase.



One maltase *enzyme* can process ***1,000 maltose molecules*** a second.

ENERGY DEPOT

One of the most important jobs that enzymes have to do is to make energy for the cell. A team of enzymes carry out a process called glycolysis, which converts glucose into new molecules. These include two molecules of a chemical called pyruvate and two molecules of an energy-rich compound called adenosine triphosphate (ATP). Some ATP is stored and the rest travels to the mitochondria with the pyruvate.



Glycolysis literally means ***"splitting sugars."***

The *human body* replaces